THE CHARUTAR VIDHYA MANDAL UNIVERSITY
A D PATEL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY

Project Title :
Expense Tracker System

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Java Source Code :-

```
package in.ac.adit.pwj.miniproject.expenses;

import java.util.*;


class Expense {
    protected String category;
    protected double amount;

    public Expense(String category, double amount) {
        this.category = category;
        this.amount = amount;
    }

    public void display() {
        System.out.println("Category: " + category + " | Amount: " + amount);
    }
}

class FoodExpense extends Expense {
    public FoodExpense(double amount) {
        super("Food", amount);
    }
}

class TravelExpense extends Expense {
    public TravelExpense(double amount) {
        super("Travel", amount);
    }
}


class ExpenseTracker {

    private List<Expense> expenses = new ArrayList<>();
```

```
    class ReportGenerator {
        public void generateReport() {
            double total = 0;

            System.out.println("\n--- Expense Report ---");
            for (Expense e : expenses) {
                e.display();
                total += e.amount;
            }

            System.out.println("Total Expense: " + total);
        }
    }

    public void addExpense(Expense e) {
        expenses.add(e);
    }

    public void showExpenses() {
        if (expenses.isEmpty()) {
            System.out.println("No expenses found.");
        } else {
            for (Expense e : expenses) {
                e.display();
            }
        }
    }
}


public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        ExpenseTracker tracker = new ExpenseTracker();
        ExpenseTracker.ReportGenerator report = tracker.new ReportGenerator();

        while (true) {
            try {
```

```
System.out.println("\n1. Add Expense");
System.out.println("2. View Expenses");
System.out.println("3. Generate Report");
System.out.println("4. Exit");
System.out.print("Enter choice: ");

int choice = sc.nextInt();

switch (choice) {
    case 1:
        System.out.println("Select Category:");
        System.out.println("1. Food");
        System.out.println("2. Travel");

        int type = sc.nextInt();

        System.out.print("Enter amount: ");
        double amount = sc.nextDouble();

        if (amount < 0) {
            throw new IllegalArgumentException("Amount cannot be
negative!");
        }

        if (type == 1) {
            tracker.addExpense(new FoodExpense(amount));
        } else if (type == 2) {
            tracker.addExpense(new TravelExpense(amount));
        } else {
            System.out.println("Invalid category!");
        }
        break;

    case 2:
        tracker.showExpenses();
        break;

    case 3:
        report.generateReport();
        break;
```

```
                case 4:
                    System.out.println("Exiting...");
                    return;

                default:
                    System.out.println("Invalid choice!");
            }

        } catch (InputMismatchException e) {
            System.out.println("Invalid input! Enter numbers only.");
            sc.nextLine(); // clear buffer
        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
}
```

Output :-

```
1. Add Expense
2. View Expenses
3. Generate Report
4. Exit
Enter choice: 1
Select Category:
1. Food
2. Travel
1
Enter amount: 10000

1. Add Expense
2. View Expenses
3. Generate Report
4. Exit
Enter choice: 1
Select Category:
```

```
1. Food
2. Travel
2
Enter amount: 15000

1. Add Expense
2. View Expenses
3. Generate Report
4. Exit
Enter choice: 22
Invalid choice!

1. Add Expense
2. View Expenses
3. Generate Report
4. Exit
Enter choice: 3

--- Expense Report ---
Category: Food | Amount: 10000.0
Category: Travel | Amount: 15000.0
Total Expense: 25000.0

1. Add Expense
2. View Expenses
3. Generate Report
4. Exit
Enter choice: 4
Exiting...
```